

SENSY REMOTE: Change TV Channels With Your Smartphone



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Nowadays, with hundreds of TV channels to choose from, it is quite a task to remember channel numbers or know what is running on which channel at a time. Mobile applications that act as TV guides are available, but these are complicated to use, especially for the elderly. To overcome this shortcoming, innovators at Sensara Technologies have introduced a device that can turn your mobile into one consolidated remote for your TV, STB and other devices. The device is called Sensy Remote.

Sensara initially developed their own TV guide mobile application called Sensy. However, soon they realised the common problem of people misplacing their remote control or using separate remotes for separate systems. Thus started their journey towards something new.

After a couple of iterations, the team launched Sensy Remote in its current avatar—an Android- and iOS-capable, battery-powered, Bluetooth remote to control TVs, STBs as well as air-conditioners. It took the team just over a year to come up with the product, which they introduced in the market early this year.

The main challenge

One of the most significant challenges during device development was to add support for STBs/remotes from the lesser known cable operators across the country. Replacement remotes for set-top boxes provided by local cable operators of different regions were hard to come by. Fortunately, help came from the common users of these STBs, who sent their

Technical features

1. 360-degree infrared coverage with 4x high-power 20kHz to 60kHz IR LED
2. Bluetooth connectivity with the smartphone
3. IR range of 4.6 to 6 metres (15 to 20 feet) with no obstruction; Bluetooth range up to 9 metres (30 feet)
4. Two AA-size batteries with expected life time of one year
5. Compatibility with over 100 DTH/cable operator STB remotes

remotes to the team. Developers utilised and integrated the pertinent services into their support database.

The science behind Sensy

Sensy Remote is a Bluetooth-driven device that lets you control your TV and STB with the help of your smartphone. It automatically connects with your mobile device via Bluetooth, and receives commands to send infrared signals to the TV or STB. You can send these commands by using a mobile application developed by the company, called the Sensy app. The Sensy application is compatible with both iOS and Android. It stores and supports data for over a hundred DTH or STB brand settings. The application sends commands to Sensy Remote through BLE (Bluetooth Low Energy, also called Bluetooth Smart).

The Sensy Remote has an integrated circuit (IC), which acts both as a microcontroller and a Bluetooth transceiver. Other major components in the circuit are a 16MHz crystal oscillator, a bulk capacitor, a p-channel MOSFET, and four LEDs that emit infrared signals to the TV or STB.

In the circuit, the Bluetooth transceiver receives commands from the mobile application. The microcontroller interprets these commands and the channel



Sensy front (top) and back (bottom)